



Program Name: MCA

COURSE PLAN: THEORY COURSE

School Name:	School of Computer Engineering			
Course Name & code:	Web Technologies MCA 4123		Core	
Semester & branch:	I semester MCA			
Name of the faculty:	Mr. Akshay Bhat, Mr. Shameem, Dr. Chetana Pujari			
No of contact hours/week:	L	T	P	C
	3	1	0	4

Course Outcomes (COs/CLOs) to PO, LO, BL Mapping#**

CO/ CLO	Statements	No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes	BL (Recom mended) *
1	Describe the semantics of HTML documentation	7	20	1, 2, 3, 8		2, 3
2	Apply the power of CSS in UI designing	10	20	1, 2, 3, 8		2, 3
3	Create interactive client-side webpages	13	25	1, 2, 3, 8		3, 4
4	Apply Server-Side data management using PHP	13	25	1, 2, 3, 8		3, 4
5	Create data-driven web pages	5	10	1, 2, 3, 8		3, 4
	Total	48	100			

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*Preferred BL is 3 or above (apply)

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
1	2							1
2	2	2	2					1
3	2	2	2					1
4	2	2	2					1
5	2	2	2					1
Average Articulation Level	2	2	2					1

Module Matrix[#]

CLO	M1	M2	M3	M4	M5	M7	M16	M17
1								
2								
3								
4								
5								
6								

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Delivery and Assessment Plan of LOs

Learning Outcome (LO) mapped to the course		Delivery and Assessment Plan
LO	LO statement	

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Assessment Plan

1. Assignment 1 (Quiz / short answer on HTML) 5 marks
2. Assignment 2 (Quiz / short answer on CSS) 5 marks
3. Assignment 3 (Surprise assessment, problem-oriented on Client-side and PHP) 10 marks
4. Mid-term exam – 30 marks (as per the template) HTML, CSS and Client-side basics
5. End-of-semester exam – 100 marks (Entire syllabus)

Lesson Plan

L No	Topics	CO Addressed
L0	Introduction to HTML, Doctype specifications, Browser Mode	CO1
L1	Anatomy of HTML elements, Document structure, well-formed HTML document	CO1
L2	Semantics of HTML Elements and Attributes, Resource path conventions, Hyperlinks	CO1
L3	HTML Table elements	CO1
L4	Web Form design	CO1
L5	HTML Form and related elements	CO1
T1	Example on HTML form	CO1
L6	Introduction to CSS, Selectors	CO2
L7	Cascading style rules	CO2
L8	CSS Box Model	CO2
L9	Positioning and Layering elements	CO2
T2	Styling contents and Images	CO2
L10	Approaches to CSS layout	CO2
L11	Responsive Designing	CO2
T3	Examples on Responsive Layout	CO2
L12	CSS Animations and Transitions	CO2
L13	Examples on animation and transitions	CO2
L14	Introduction to DHTML	CO3
L15	Data Types and Objects in JavaScript	CO3
T4	Event Handling and Error Handling in JavaScript	CO3
L16	Cookies	CO3
L17	Handling Cookies	CO3
L18	Introduction to Form Validation	CO3
L19	Regular Expression	CO3
T5	Examples on Regular Expression	CO3
L20	Client-side form handling	CO3
L21	Client-side form handling	CO3
L22	Introduction to AJAX	CO3
T6	Examples on Client-side AJAX	CO3
L23	Canvas application	CO3
L24	Introduction to client-server architecture	CO4
T7	Server-side programming using PHP	CO4
L25	PHP controls and functions	CO4
T8	Examples on basics	CO4
L26	Arrays and superglobals	CO4
L27	Examples on Arrays and super global	CO4
L28	File Handling	CO4
L29	Server-side form validation	CO4
T9	Example on Server-side form validation	CO4

L30	Introduction to server-side state management	CO4
L31	Cookies	CO4
T10	Session	CO4
L32	Exception handling	CO4
L33	MySQL with PHP	CO5
T11	Data Operations with PHP	CO5
L34	CRUD Examples	CO5
L35	Data Handling with JSON	CO5
T12	Examples JSON	CO5

Self Directed Learning (SDL)

Sl. No.	TOPIC (Research Literature/ Case Studies NPTEL/Coursera/any MOOC-based material Virtual Lab, etc.)	No. of learning hours	CO
1	Angular Dependency Injection Components, Templates, Binding , Forms	72	CO5
	Total number of learning hours	72	

***The SDL topics may be from the Research Literature/ Case Studies NPTEL/Coursera/any MOOC-based material Virtual Lab, etc.**

****One credit is equivalent to 30 hours of learning.**

References:

Textbooks	1.	Randy Connolly, Ricardo Hoar, Fundamentals of Web Development, 1st Edition, Pearson Education India, 2015.
	2.	Luke Welling, Laura Thomson, PHP and MySQL Web Development, 5th Edition, Pearson Education, 2016
	3.	Nicholas C Zakas, Professional JavaScript for Web Developers, 3rd Edition, Wrox/Wiley India, 2012.
	4.	John Kocer, “Angular 7: By Example (Part One Book 1)”, 2019.
	5.	Nate Murray, Felipe Coury, Ari Lerner, Carlos Taborda, ng-book- The Complete Book on Angular, 2019.
Self-Directed Learning		
Research Literature/ Case Studies		
Coursera		Mastering Angular: Foundations to Enterprise-Ready Apps Specialization
Virtual Labs		Angular.dev (online lab)

Submitted by:

(Signature of the faculty)

Date:

Approved by:

(Signature of the Dean)

Date:

Faculty members teaching the course (if multiple sections exist):

Faculty Name	Section	Signature	Faculty Name	Section	Signature
Mr. Akshay Bhat	A		Dr. Chethana Pujari	C	
Mr. Shameem	B				